CityCycle London Bike Rebalancing Intelligence Pipeline

Team 2

Overview

01 Introduction

The problem, proposed solution, business impact.

02 Data Strategy

Architecture, dataflow diagram, analysis, modeling.

03 Key Insights

Signals from the data, imbalance analysis, action steps.

04 Risks & Mitigation

Pipeline risks, model failures, mock-first approach

05 Demo

Dashboard, Looker Studio, Notebooks

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01
Introduction

The Problem

London's CityCycle network struggles with chronic bike imbalance across 795 docking stations — stranded riders, overflow docks, and reactive crew operations.

01

Empty Stations

Riders arrive to find no bikes during peak commute hours. Lost rentals, frustrated customers, negative NPS.

02

Overflowing Docks

Returning riders cannot dock bikes — no empty slots. Bikes abandoned nearby, increasing theft and damage risk.

03

Reactive Rebalancing

Crews deploy reactively based on reports, not data. Inefficient routing wastes time, fuel, and crew hours.

04

No Demand Forecasting

No predictive model for station-level demand. Operations cannot pre-position bikes before peak periods.

The Proposed Solution

01

Ingest

32M rides from BigQuery public dataset via Meltano EL pipeline

02

Transform

dbt ELT - 7 models, star schema, derived imbalance metrics

03

Quality Check

Great Expectations - 30 PASS · 4 WARN · 0 FAIL across 34 checks

04

Analyse & Serve

EDA with Jupyter Notebook + SQL Alchemy
ML Training
Streamlit Intelligence Dashboard
Looker Studio Visualisation

05

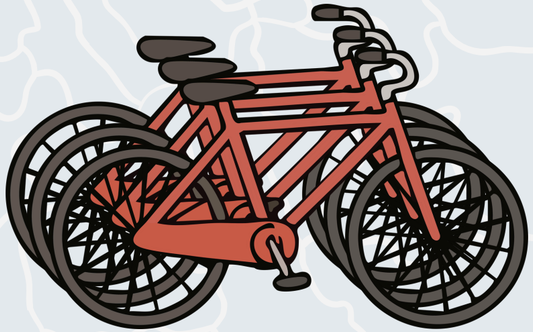
CI Github Actions

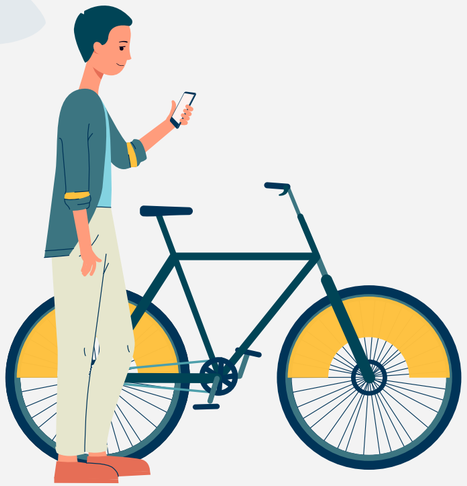
To verify the codebase is healthy with every push to github.

Business Impact

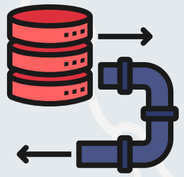
Data-driven rebalancing transforms CityCycle from **reactive to predictive operations**.

From the analysis we will be able to identify usage trends and station imbalances, leading to intelligent crew routing that reduces costs, prevents empty docks, and protects revenue - built on a **reusable ELT architecture that scales to any city**.

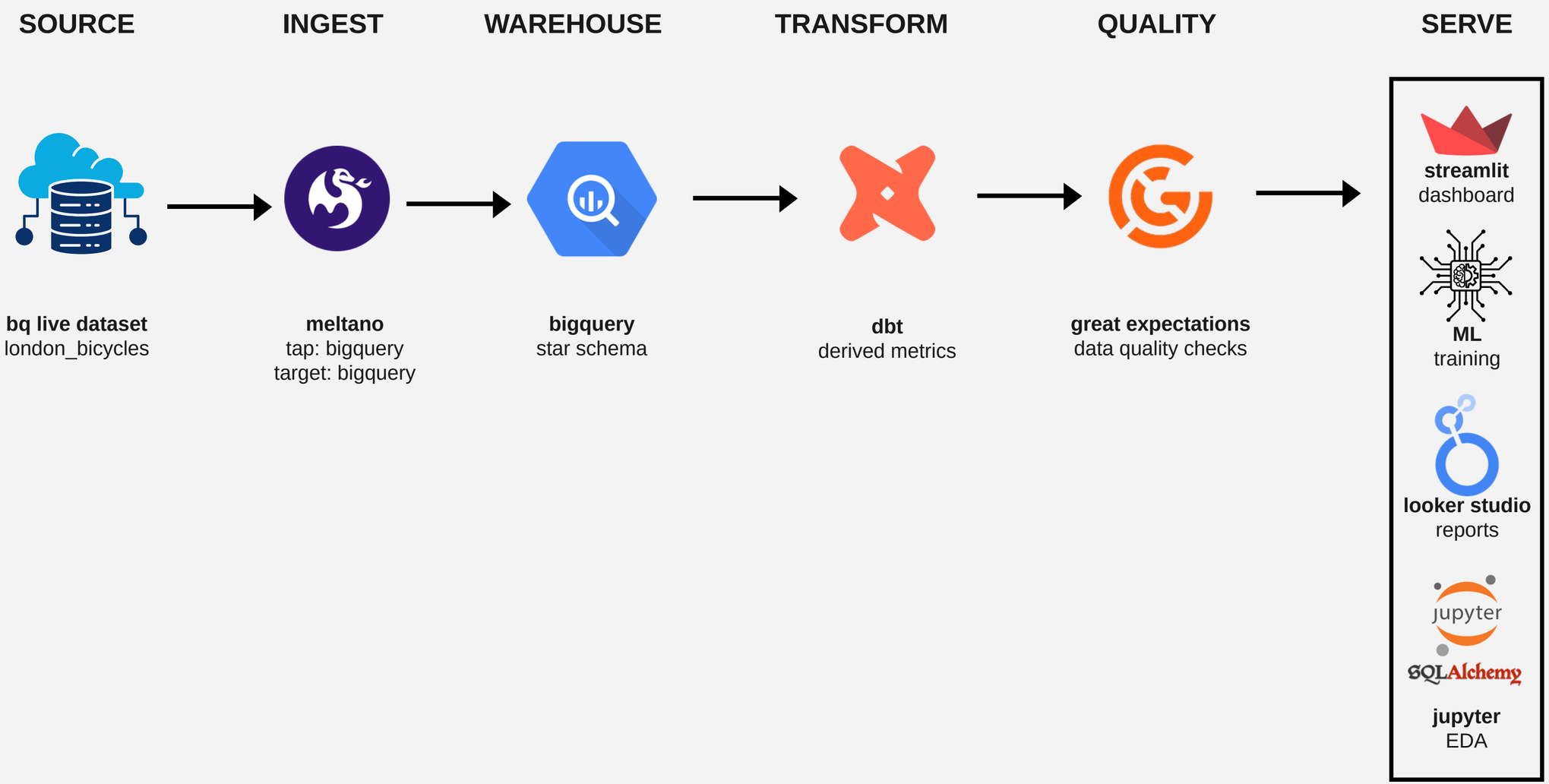




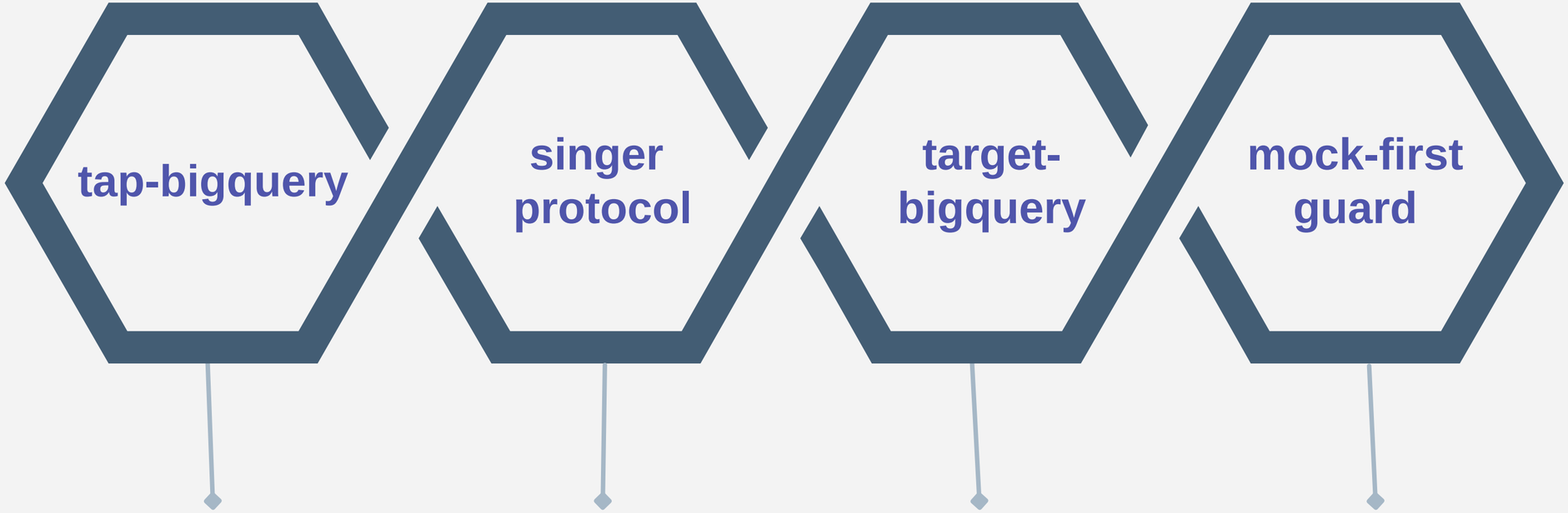
02
Data Strategy



Data Flow Diagram



01 - Data Ingestion: Meltano EL Pipeline



Singer tap reads from bigquery-public-data.london_bicycles. Supports full refresh and incremental loads using start_date state key. Schema auto-detected.

Structured streams of CATALOG, STATE and RECORD messages. Type-safe column mapping enforced. Compatible with any Singer target.

Loads to citycycle_raw dataset in EU region. 32M rows in cycle_hire, 795 rows in cycle_stations.

All development uses 10K synthetic rows — zero BQ scan cost in CI. Dry-run cost estimates before every live query. 1TB/month free tier protected.

02 - Transformation: dbt Pipeline

STAGING

stg_cycle_hire · stg_cycle_stations

Materialised as View

Derived Fields

hire_date
start_hour
day_of_week
is_weekend
duration_seconds
zone
capacity_tier

INTERMEDIATE

int_rides_enriched · int_station_daily_stats

Materialised as View

Derived Fields

duration_minutes
duration_band
peak_hour_flag
time_period
is_round_trip
net_flow
imbalance_score
utilisation_rate
is_imbalanced
imbalance_direction

MARTS

fact_rides · dim_stations · dim_date

Table, Partitioned, Clustered

Derived Fields

ride_sk(surrogate key)
rolling_7d_avg (ML feature)
start_station_imbalance_score
start_station_net_flow
start_station_is_imbalanced
rebalancing_priority
avg_imbalance_score_7d
season
is_uk_bank_holiday

Source: BigQuery

Streamlit · Looker Studio · XGBoost

dbt 1.x · BigQuery adapter · dbt_utils package · 56 tests PASS · 0 ERROR · 3 WARN (intentional severity: warn)

02 - Data Warehouse: Star Schema

A star schema optimises analytical queries — one central fact table surrounded by dimension tables. This structure supports both Streamlit and Looker Studio without complex joins across large tables.

dim_stations (798 rows)

station_id (PK)
station_name
zone
latitude
longitude
nb_docks
capacity_tier
rebalancing_priority
avg_imbalance_score_7d
is_installed
is_locked

fact_rides (32.3M rows)

rental_id
ride_sk (PK)
hire_date (partition key)
start_station_id (FK)
duration_minutes
peak_hour_flag
imbalance_score 0–1
imbalance_direction
net_flow
rebalancing_priority

dim_date (4000 rows)

date_id (PK)
full_date
year
month
day
week_num
day_of_week
is_weekend (Bool)
season (spring/summer/…)
Date spine: 2010–2030

dbt Transformation Pipeline · 7 models · 56 tests PASS · 0 ERROR · 3 WARN

Staging (clean + type)
stg_cycle_hire
stg_cycle_stations

→

Intermediate (join + enrich)
int_rides_enriched
int_station_daily_stats

→

Marts (star schema)
fact_rides
dim_stations
dim_date

04 - Data Quality: Great Expectations

30	4	0	34
PASS	WARN	FAIL	TOTAL

Checkpoint 1 — Post-Ingest (citycycle_raw)	
PASS	rental_id — not null + unique
PASS	start_date — after 2010-01-01
PASS	start_station_id — not null
PASS	latitude/longitude — London range
PASS	total row count — at least 1M rows
WARN	duration < 60s — 27K docking errors (source data)
WARN	end_station_id NULL — 312K lost/unreturned bikes
WARN	end_date ≤ start_date — 10K timestamp errors (source)

Checkpoint 2 — Post-Transform (marts)	
PASS	ride_sk surrogate key — unique
PASS	duration_minutes — between 1 and 1440
PASS	start_hour — between 0 and 23
PASS	duration_band — valid values only
PASS	null rate on key columns — under 5%
PASS	end_datetime > start_datetime
PASS	imbalance_score — between 0 and 1
WARN	FK: start_station_id → dim_stations (hist. stations)

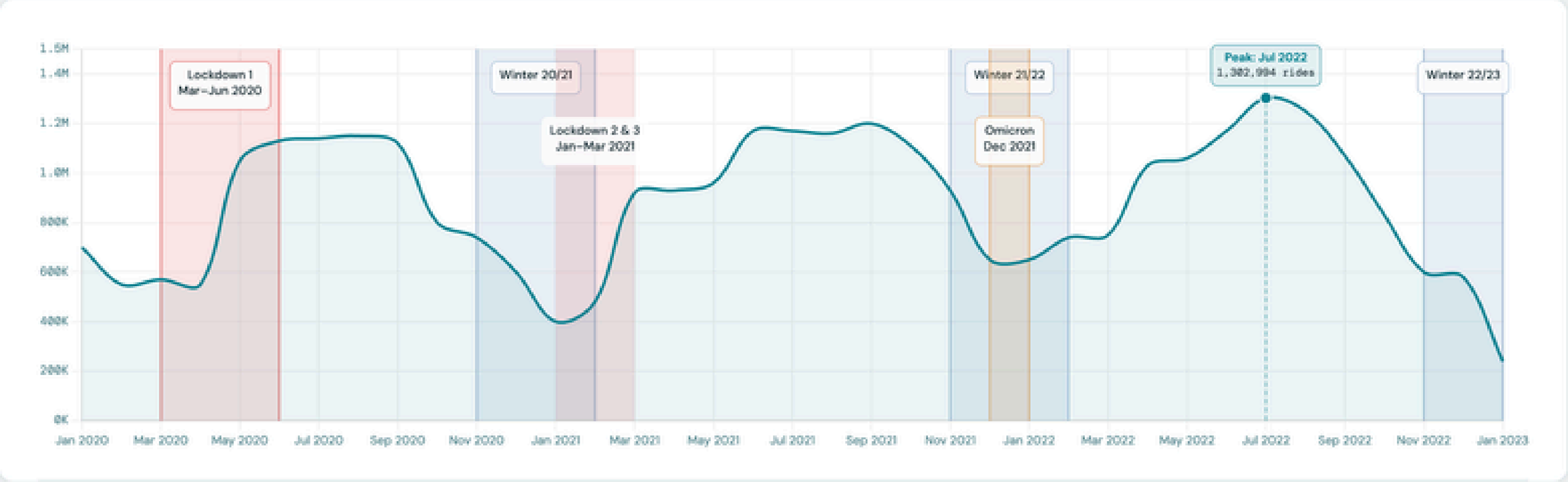
03
Key Insights

Key Findings

Metric	Value	Insight & Business Implication
Total rides (2020–2023)	32,342,086	3 years of real operational data — sufficient for trend detection and ML model training
Average ride duration	21.8 minutes	Mixed use confirmed — commuter and leisure trips coexist across the network
Rider segments	Leisure 53% · Casual 32% · Commuter 15%	K-Means k=3 — each segment drains stations in different directions at different times
Weekend rides	9.47M (29.3%)	Strong leisure demand — different rebalancing strategy required Sat/Sun vs weekdays
Avg network imbalance score	0.084	Network-wide baseline — scores above 0.18 trigger scheduled crew intervention
Critical stations	3 stations	Score ≥ 0.25: New North Road Hoxton · Ladbroke Grove Central · Cloudesley Road Angel
High priority stations	27 stations	Score ≥ 0.18 — require scheduled daily intervention by rebalancing crews
XGBoost RMSE	2.422 rides/hr	Best of 3 models tested — within operational planning threshold for crew dispatch
Top draining station	New North Road Hoxton	Score 0.324 · Net flow +7.0 bikes/day · Draining 90% of days — needs daily pre-AM crew
COVID-19 signal	Visible in 2020	Clear ridership collapse March–May 2020, full recovery to pre-pandemic levels by mid-2022

Ride Trends 2020 – 2023

CityCycle London · Monthly ride volume · 32,342,086 rides · bigquery-public-data.london_bicycles via Meltano → dbt → BigQuery



Monthly rides · COVID lockdown periods · Winter seasonality (Nov–Feb) · Omicron / Plan B (Dec 2021)

Source: citycycle-dsai4.citycycle_dev_marts.fact_rides · DSAI4 Module 2 · Team 2 · March 2026 · Monthly values approximated from Looker Studio. Jul 2022 exact: 1,302,994 rides (notebook Cell 11).

Understanding Imbalance

Imbalance Score = ABS(departures − arrivals) ÷ (departures + arrivals)

Range: 0.0 (perfectly balanced) → 1.0 (entirely one-directional) · Calculated daily per station ·CRITICAL threshold: 0.25

Filling

net_flow < 0

More bikes arriving than leaving.
Docks filling toward overflow.

Crew Action:
COLLECT BIKES — remove bikes to free up empty docking slots.

Hop Exchange, The Borough
Score 0.15 · Net flow −38/day

Balanced

net_flow ≈ 0

Departures ≈ arrivals.
Station self-regulating.

Crew Action:
MONITOR ONLY — no immediate intervention required.

Score < 0.10
Low priority station

Draining

net_flow > 0

More bikes leaving than arriving.
Station heading toward empty.

Crew Action:
DELIVER BIKES — bring bikes from nearby filling stations.

New North Road Hoxton
Score 0.324, net flow +7.0/day

Why 0.25 = CRITICAL: A score of 0.25 means 25% more bikes moving one direction than the other. At this rate, a busy station risks running empty or reaching full capacity within a single day.
Thresholds were calibrated from the all-time score distribution: CRITICAL ≥ 0.25 · HIGH ≥ 0.18 · MEDIUM ≥ 0.10 · LOW < 0.10

04

Risk Mitigation

Decisions & Risk Management

KEY RISKS & MITIGATIONS

- **BQ free tier exceeded (1TB/month)**
Dry-run cost checks before every query · LIMIT guards · date partitioning · budget alert at 80%
- **Source schema drift (TfL updates)**
dbt schema tests catch column changes on next run · GE post-ingest checkpoint adds second layer
- **ML model staleness**
Retrain script fully rerunnable
- **Source data quality (public dataset)**
312K null end_station_id · 27K short durations · 10K timestamp errors — all flagged as GE WARN not FAIL
- **Credentials leaked to Git**
.gitignore blocks all credential patterns · .env.example only committed · profiles.yml excluded

FAILURES ENCOUNTERED & FIXED

- **BOOL/INT64 type mismatch in BigQuery**
COUNTIF(col) replaced with SUM(CAST(col AS INT64)) across all queries, notebooks and dashboard pages
- **dim_stations all showing LOW priority**
7-day WHERE filter in latest_stats CTE returned zero matches · Fixed with all-time averages · Thresholds recalibrated
- **XGBoost feature order mismatch**
season column in wrong position between training and dashboard · Explicit column reorder enforced in CI and dashboard
- **APPROX_QUANTILES breaks SQLAlchemy**
Returns ARRAY type which SQLAlchemy cannot deserialise · Replaced with AVG throughout

MOCK-FIRST DEVELOPMENT

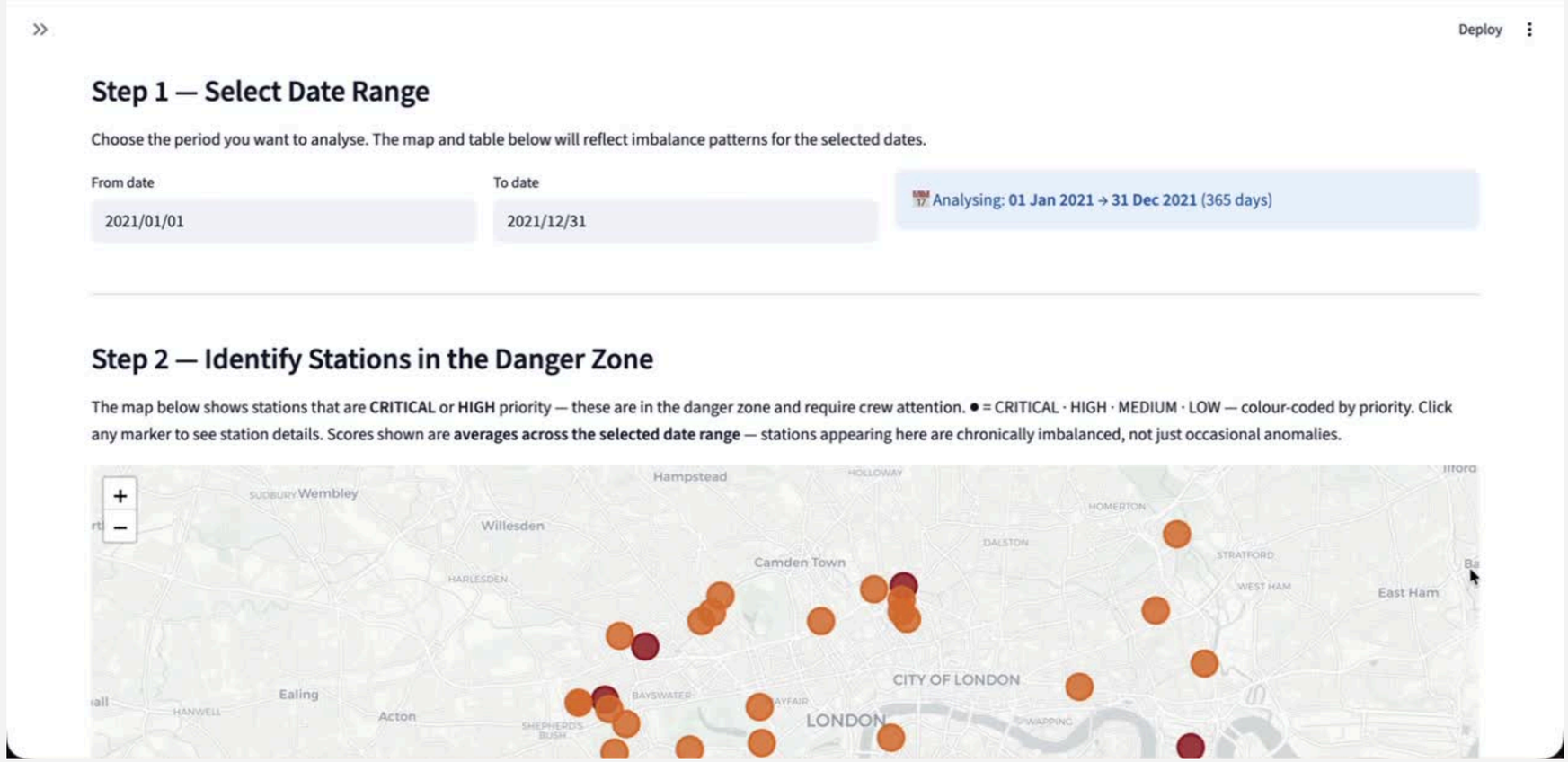
- **Why mock-first?**
london_bicycles has 83M rows — one unguarded SELECT * consumes the entire 1TB monthly free tier instantly
- **10K synthetic rides (numpy, seed=42)**
Schema matches real BQ tables exactly - same columns, types and constraints.
- **CI/CD runs zero live BQ calls**
All 5 GitHub Actions jobs use mock CSV only · dbt compile validates SQL at zero cost · No credentials in CI
- **Dev toggle in dashboard**
Streamlit use_mock switch — one click to test against mock before hitting live BQ
- **Outcome: $0 development cost**
Entire pipeline built and validated without triggering BQ billing · Live run completed safely in one production pass

05
Demo

Scenario Planner

COMMAND:
streamlit run
dashboard/app.py

Audience: Operations teams need interactivity and real-time filters.
Purpose: Interactive exploration and crew dispatch planning



Demo Links

Looker Studio:
https://lookerstudio.google.com/s/hNwE4QyCuCg

GitHub Repo:
https://github.com/berniepng/dsai4-m2-t2-citycycle

Streamlit:
streamlit run dashboard/app.py, OR open ~/docs/dashboard/05_scenario_static.html

Jupyter Notebook:
open ~/analysis/notebooks/03_bq_eda_live_data_executed.ipynb in IDE

06

Questions?